
Boston University MET CS 581

Introduction to Health Information Systems

Healthcare Workflow
Business Process Re-engineering

Michael Levinger

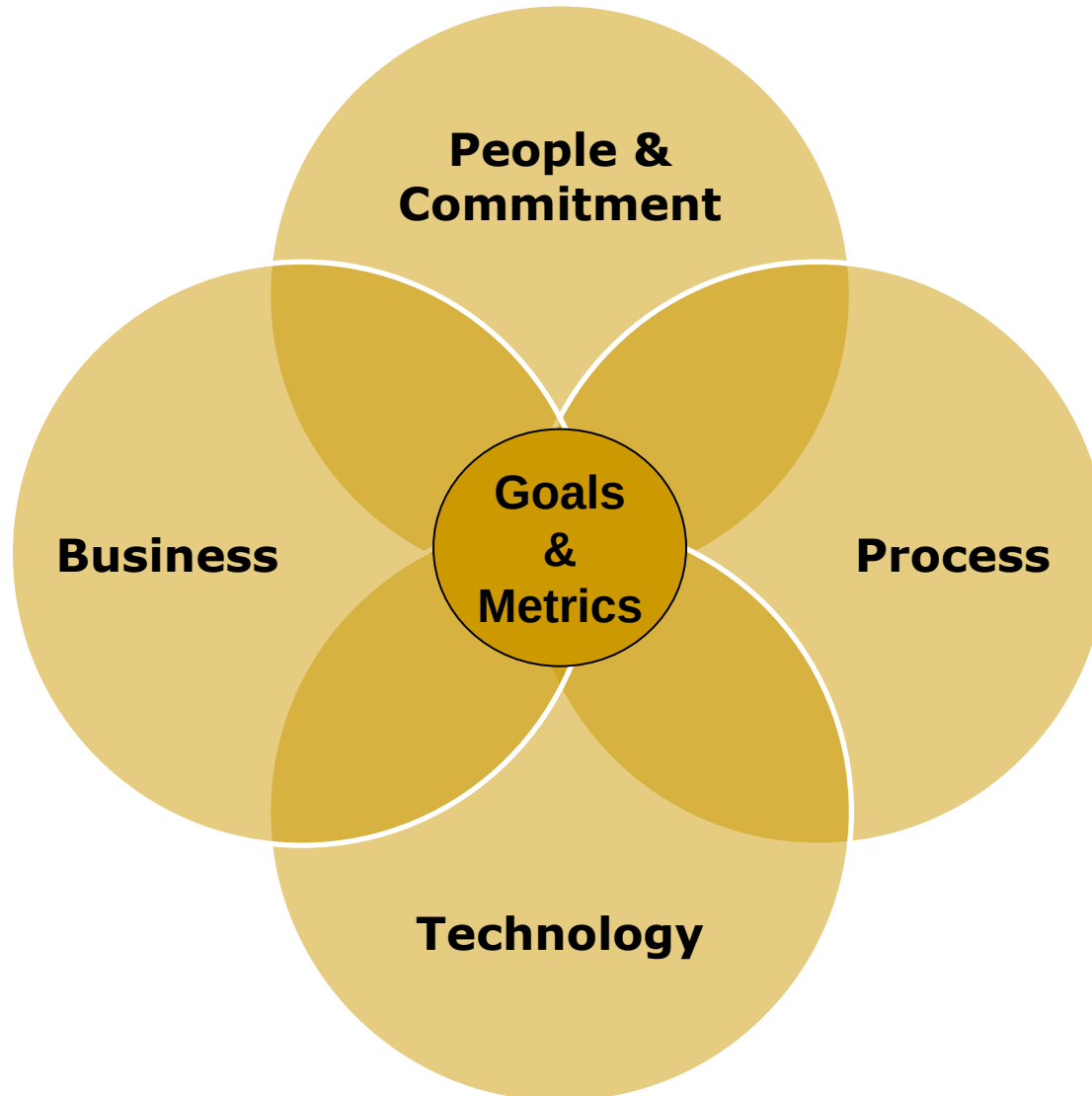
January 15, 2025

Agenda

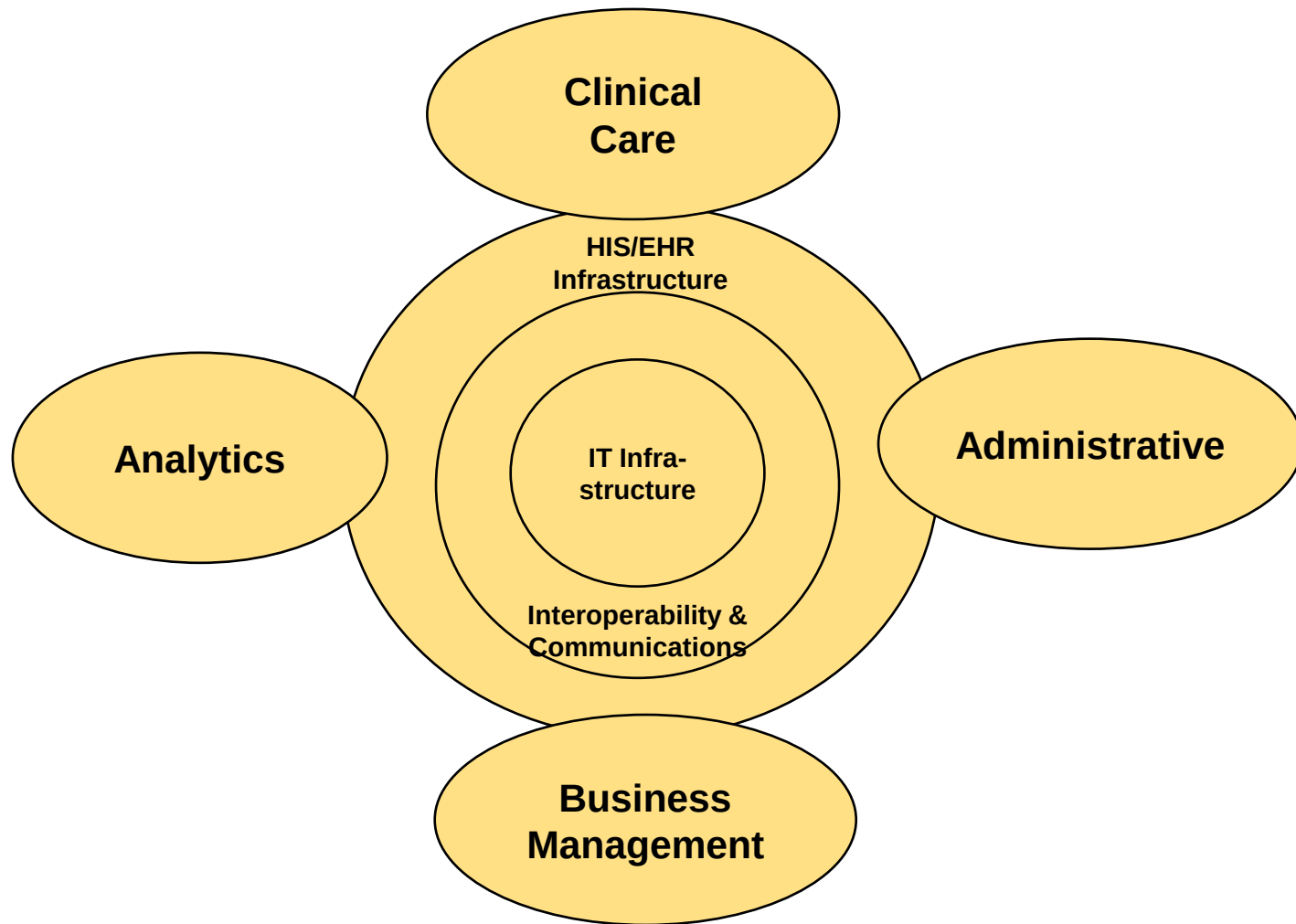
- Short CS581 Course Introduction
- Overview of Clinical Workflow
- Overview of Business Process Re-engineering (BPR)
- Implementing an HIS/EHR – Impact on Clinical Workflow

HIS/EHR Success Factors

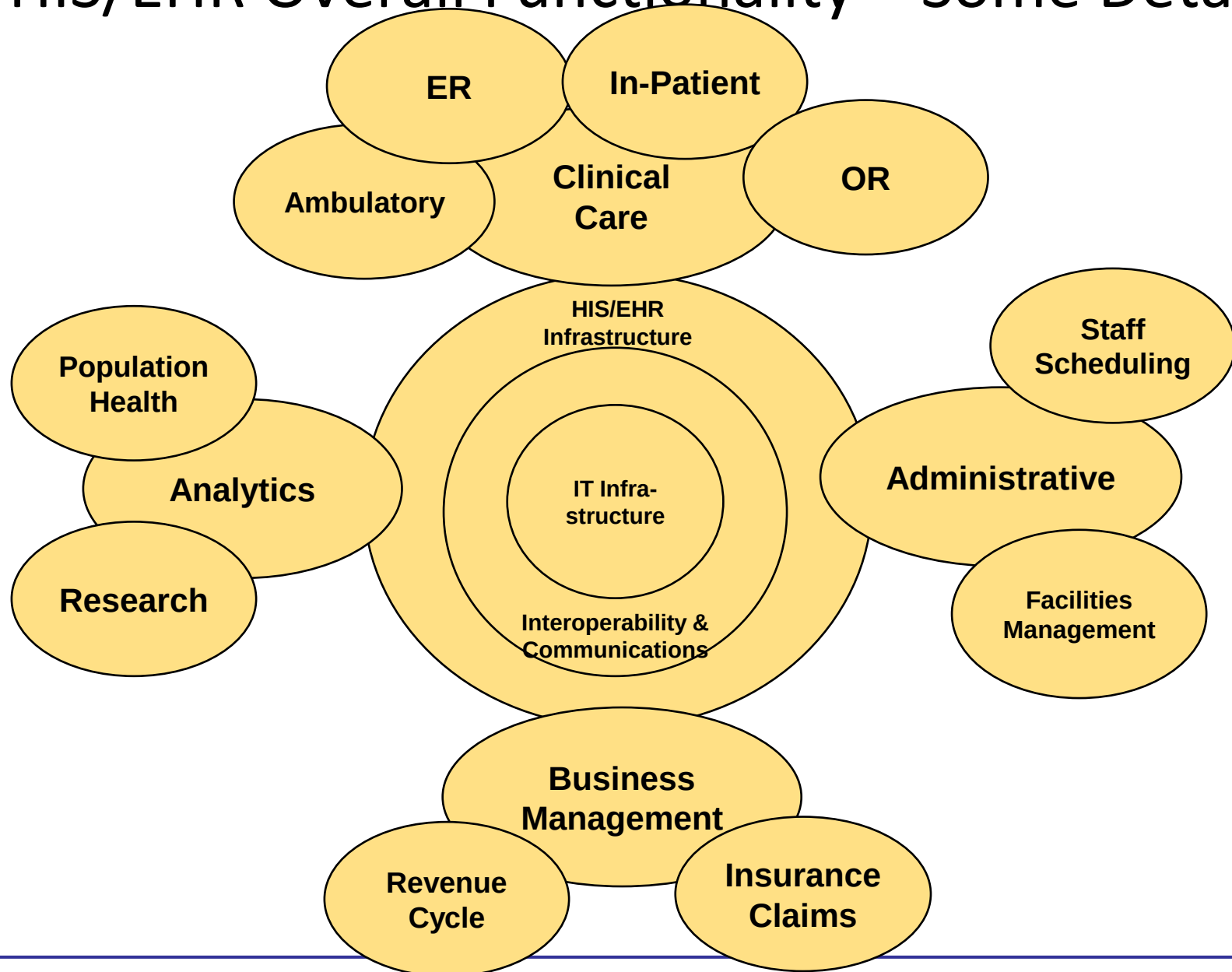
Balanced Performance Model



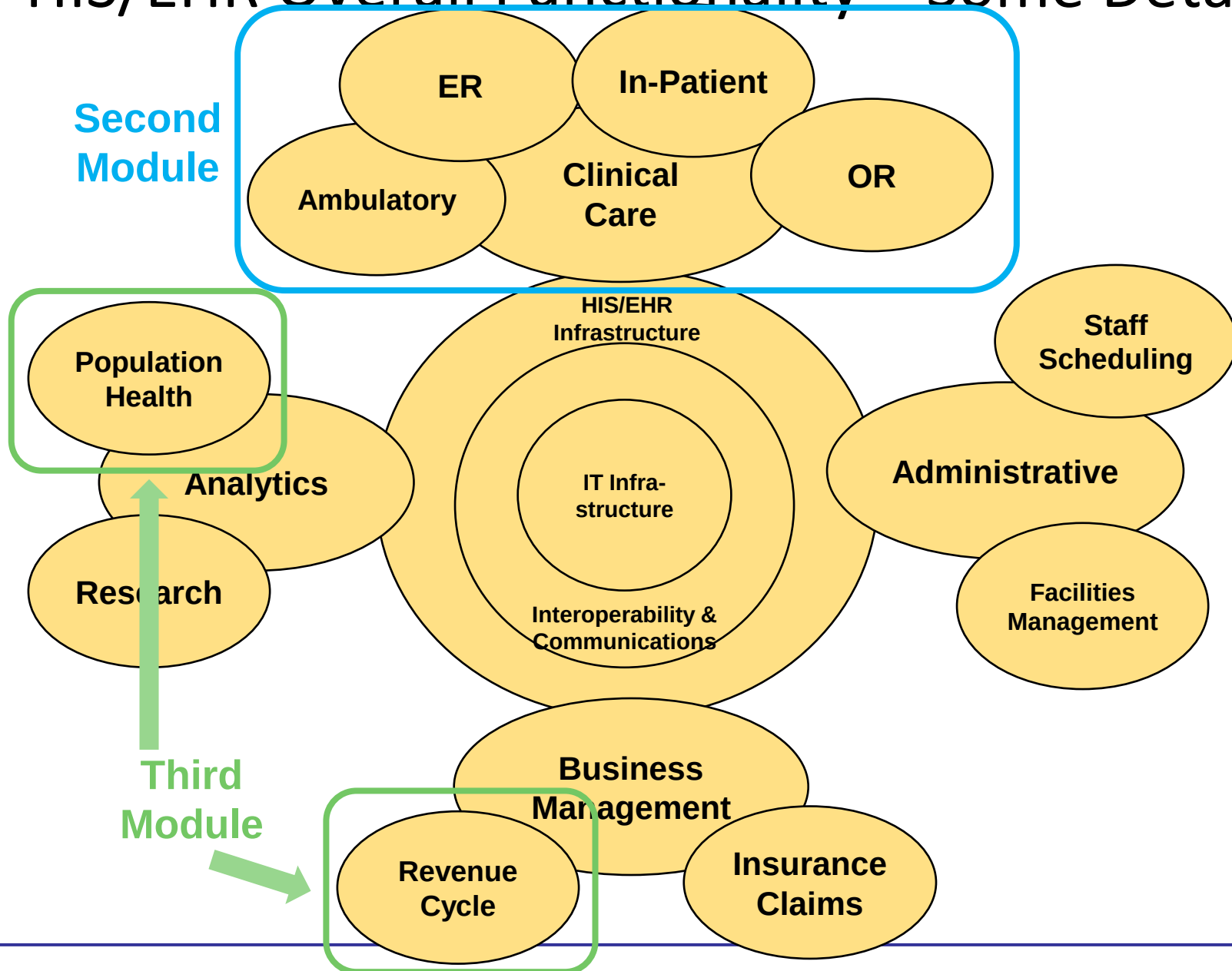
HIS/EHR Overall Functionality



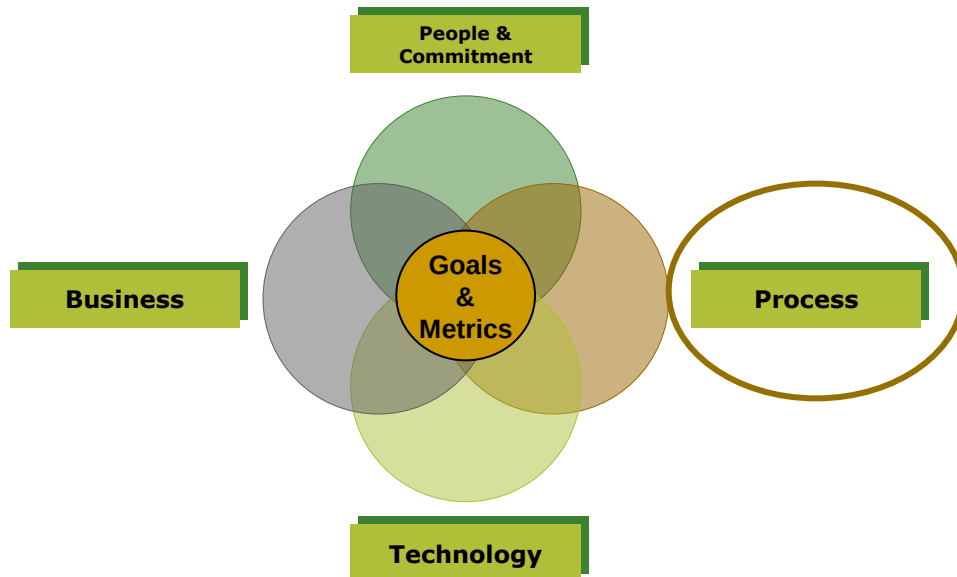
HIS/EHR Overall Functionality – Some Detail



HIS/EHR Overall Functionality – Some Detail



HIS/EHR & Healthcare Processes



- Start with What Clinicians Do
- Then How Does an HIS/EHR Impact What They Do
- Then How What Does an HIS/EHR Do
- And How Does an HIS/EHR Improve Workflow

Healthcare Clinical Processes

workflow: n., in an office or industrial organization, the sequence of processes through which a piece of work passes from initiation to completion - Oxford English Dictionary

Overview of Business Process Re-engineering

Don't Overlook Processes and People Before Implementing Technology Solutions

**Presented at
HIMSS 10th Annual Conference
Atlanta, Georgia
March 4, 2010**

By

Deborah L. Smith, Ph.D., M.B.B

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and

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Principal, Workflow IT Solutions, San Ramon, CA

1

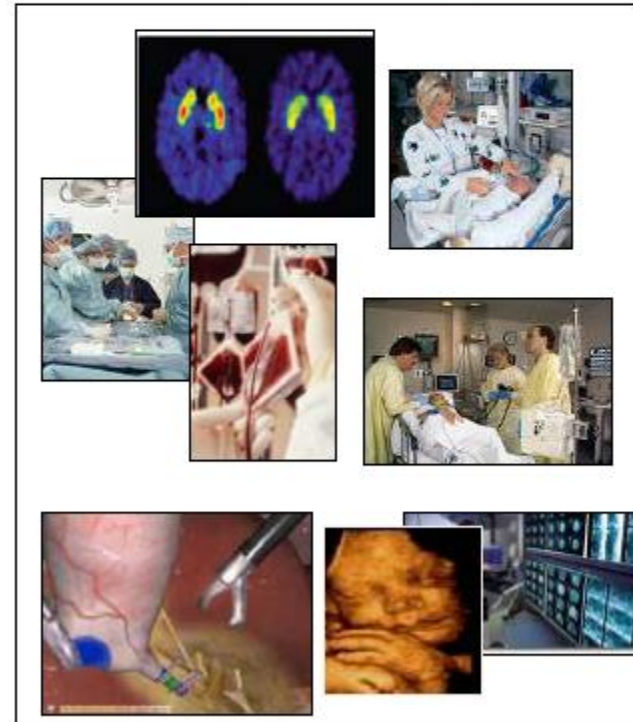
Healthcare is an Organizational System with Complex Embedded Processes

Yesterday:
The Practice of Medicine



HIMSS 2010 Atlanta, GA Processes & People by
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Today:
The Practice of Medicine



7

Importance of Improving Process Workflow



Healthcare industry is under pressure to work better, faster and leaner

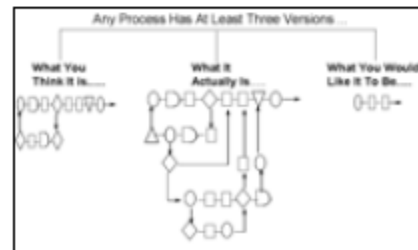
- A process map is the **blueprint** to your workflow
- When you can see your processes, you can identify shortfalls, gaps or delays before adverse events occur
- Leaders who know their processes can better manage results
- To the patient, process mapping is invisible
- Consistent processes allows better operations
- Building an infrastructure enables consistent delivery of care
- A transformation process, can hardwire the experience through process mapping

Tell me and I will forget; show me and I may remember; involve me and I will understand. Chinese Proverb

8

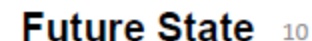
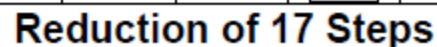
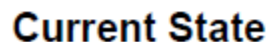
Importance of Improving Process Workflow

- When you think about it, every process has three versions:
the perceived process ("what we think it is");
the reality process ("what the process actually is")
the ideal process ("what the process could be").
- Make sure the process map you draw reflects reality.
- This is the only way you will be able to document accurate process steps.
- Regardless of the type of process map, there are several basic steps involved in creation.



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Smith & Ganti

Workflow (Cont'd)



An Example Process

Small Group Discussion

Doing Your Laundry

What is the

- Perceived As-Is Process,
- Real As-Is Process, and
- Should Be Process?

An Example Process

Small Group Discussion of the process of Doing Your Laundry

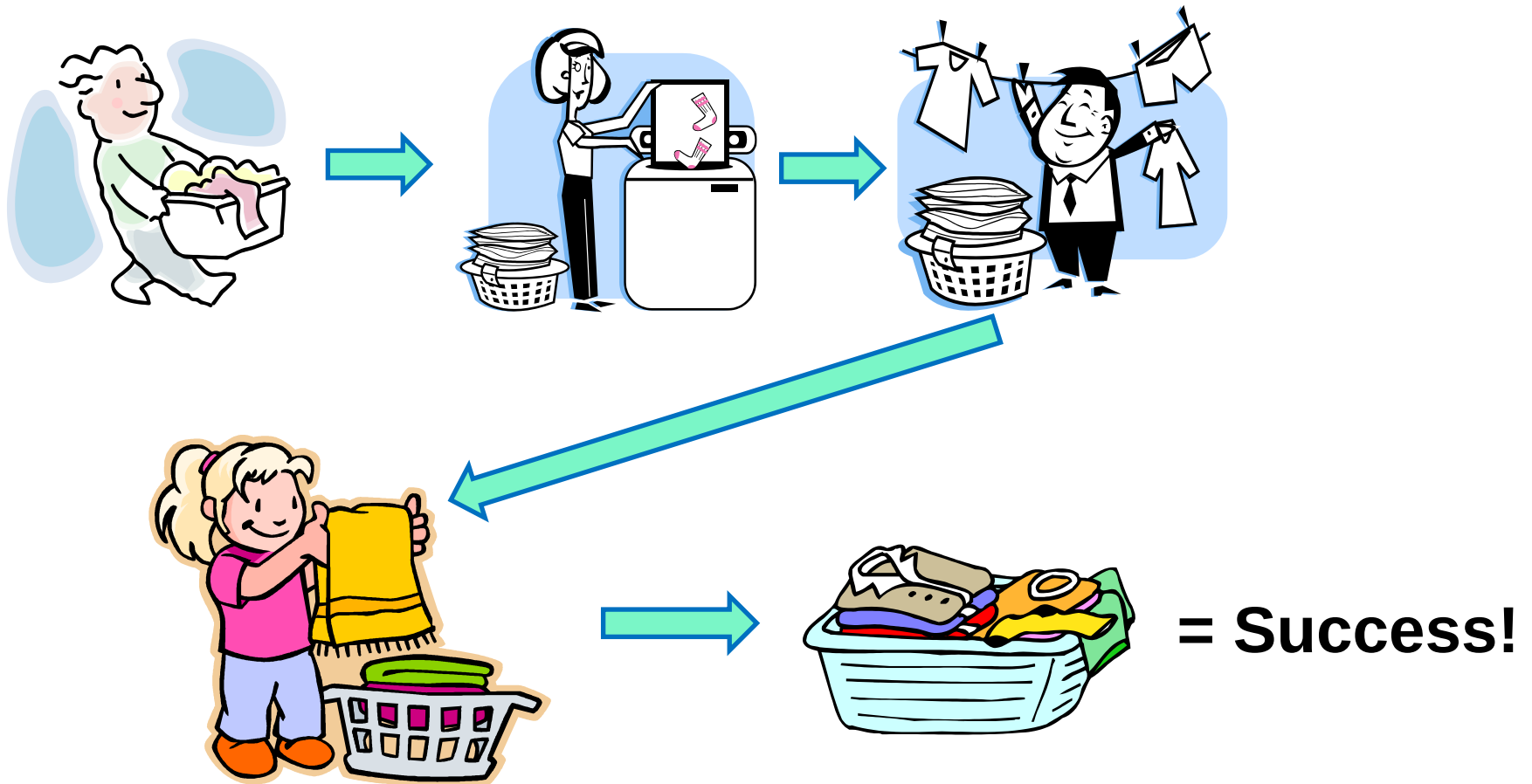
What is the

- Perceived As-Is Process,
- Real As-Is Process, and
- Should Be Process?

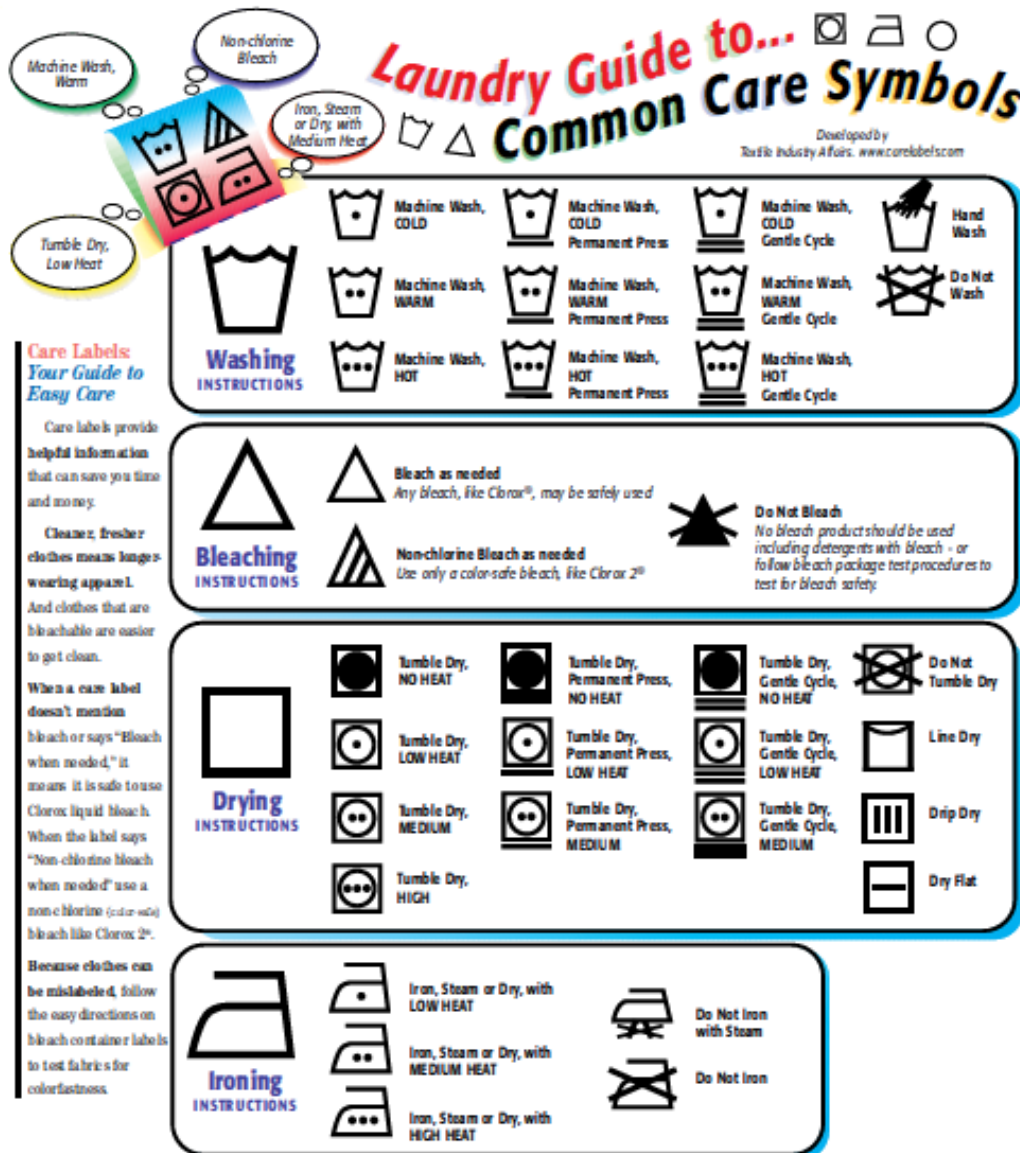
Process

1. The professor will put students into small groups
2. The groups will have 5 minutes to answer the questions
3. At the end of the time, post your group's consensus answer into the meeting Chat
4. The professor will then review the answers

Laundry Process



Laundry Process



- What is what we're told we should do?
- What do we really do?
- What should we do?

What are the benefits of doing BPR as part
of implementing an HIS/EHR?

Why Understand the Importance of Improving the Workflows Prior to Automation?

Consequences:

Increased cost due to the hidden or unknown factors=\$\$\$

More time investment = \$\$\$

Turnover of staff=\$\$\$

Possibility of a mistake or error= \$\$\$





Why Learn the Steps Involved in Mapping and Analyzing the Current State Before Developing the Desired State?

- Knowing the exact steps that have to be performed is crucial to understanding the process.
- Knowing the order of the steps in sequence or in parallel, which can help improve the overall time required to fulfill a process.
- Knowing the responsibilities, to identify accountability.
- Knowing the utilization of resources consumed in the process.
- Knowing who uses which resources can help improve the utilization of resources as resource requirements can be planned.



Why Learn the Steps Involved in Mapping and Analyzing the Current State Before Developing the Desired State?

- Knowing the relationship between people involved in the processes, and their communication.
- Knowing exactly who communicates with whom is important and can help to organize and optimize communications.
- Knowing where the documents are going, and where they are coming from is important to understand the document or information flow. Processes produce and consume information (regardless of paper or electronic documents).. A good overview of the documents also gives us the opportunity to identify whether all of the documents are really necessary. To identify potential bottlenecks and points of improvements, which can be used later in the process optimization phase. Use defined metrics for tracking.
- To improve the understandability of quality and safety regulations.

Healthcare Processes

What are major processes in healthcare?

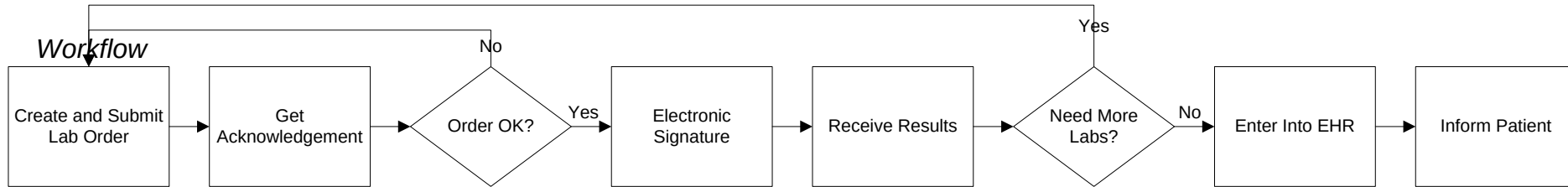
	Ambulatory	Acute	Post Acute	Clinics	Retail	Home
Clinical						
Analytic						
Business						
Admin						

Diagnostic Test Ordering – Example BPR

1. What is the workflow? (6-12 steps)
2. What is key information (3-6 items) needed at each key process step?
3. What people in the practice are involved for each process step? (3-6 people)
4. What are examples of key EHR functionality (3-6 functions) used for each key process step?
5. What are possible benefits (3-6 benefits) from changing/improving the workflow?

Diagnostic Test Ordering – Example BPR

Workflow



Data Needed

•Diagnosis •Possible labs	•Lab acknowledge ment	•Original Lab Order •Lab Ack. •Diagnosis	•Physician ID	•Original Lab Order •Lab Results •Diagnosis	•Original Lab Order •Lab Results •Diagnosis	•Lab Results •Diagnosis	•Patient Info •Lab Results •Diagnosis
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Who is involved?

•Physician •Nurse •Lab	•Physician •Nurse •Lab	•Physician •Nurse	•Physician	•Physician •Nurse •Lab	•Physician	•Physician •Nurse •Admin Staff	•Physician •Nurse •Patient
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HIS/EHR Functionality

•Order Sets •Order Protocols •Lab System (LIMS) Interface	•Lab System (LIMS) Interface	•Clinical Decision Support •Order Protocols	•Electronic Signature	•Lab System (LIMS) Interface	•Clinical Decision Support •Order Protocols	•Lab System (LIMS) Interface	•Patient Communication
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Benefit

•Faster •Less Error Prone	•Faster •More available	•Faster •Less Error Prone	•Safer •Faster	•Faster •Greater availability	•Faster •Less Error Prone	•Faster •Greater availability	•Faster •Greater availability •Improved Communication
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Assignment 1 – Workflow Redesign

Redesigning the workflow was one critical part of the effort to implement a new EHR at Virginia Women's Center (VWC). You have been asked by your manager, the CIO at Virginia Women's Center, to take the lead on process redesign as part of the EHR deployment project you are running. You are going to start with one area of clinical activity, referral to a specialist.

1. Think through the process steps required to do a referral to a specialist.
2. Develop a workflow diagram for the referral process for Virginia Women's Center. Detailed information on VWC is in the updated VWC HIMSS Davies Award application on the course site under Assignment 1.
3. Elaborate on that workflow diagram using the five-step business process re-engineering methodology outlined in Module 1 of the online course materials.
4. Identify the critical issues to successfully deploy the new workflow as part of deploying an EHR.

Assignment 1 – Workflow Redesign - Submissions

Your submission should include:

- Workflow Diagram
- What is the key information (3-5 items) needed at each key process step?
- What people are involved for each process step?
- What HIS/EHR capabilities are needed to implement the workflow?
- What are possible benefits (4-7) from changing/improving the workflow?
- Identify the critical issues to successfully deploy the new workflow as part of deploying an HIS/EHR

Submit your workflow diagram using either Microsoft PowerPoint, Microsoft Visio, Google Slides or Adobe PDF and the critical issues in in Microsoft PowerPoint, Microsoft Word, Google Docs, or Google Slides. Bullet points, diagrams etc. are fine. You do not need full written sentences.

Please submit your homework on the class Blackboard Site.

Workflow Redesign

Grading Criteria

Learning Objective	Points
Understanding of Workflow Mapping • Basic workflow • Decision Points, Branching, Repetitive Steps	30 Points
Understanding of workflow redesign and critical issues to deploy new workflow	30 Points
Understanding of “flow” from workflow to data required, people involved, EHR functionality and benefits	30 Points
Clarity and quality of the material, citations, use of AI documented	10 Points
TOTAL	100 Points